

DAX Formulas Reference

Customer Insight Analysis — Power BI Dashboard by YSG

PAGE 1 — Executive Summary

Measure: Total Customers

```
Total Customers = COUNTROWS(Sheet1)
```

Measure: Total Revenue

```
Total Revenue = SUM(Sheet1[TotalSpend])
```

Measure: Avg Satisfaction

```
Avg Satisfaction = AVERAGE(Sheet1[SatisfactionScore])
```

Measure: Churn Rate

```
Churn Rate =  
DIVIDE(  
COUNTROWS(FILTER(Sheet1, Sheet1[ChurnStatus] = "Yes")),  
COUNTROWS(Sheet1),  
0  
) * 100
```

Charts used: KPI Cards × 4, Clustered Bar (Revenue by Region), Donut Chart (Customers by Segment)

PAGE 2 — Customer Behavior

Calculated Column: Age Group

```
Age Group =  
SWITCH(  
TRUE(),  
Sheet1[Age] < 30, "18-29",  
Sheet1[Age] < 40, "30-39",  
Sheet1[Age] < 50, "40-49",  
"50+"  
)
```

Create via: Modeling → New Column

Measure: Avg Spend

```
Avg Spend = AVERAGE(Sheet1[TotalSpend])
```

Measure: Total Revenue (used in Top Product Categories)

```
Total Revenue = SUM(Sheet1[TotalSpend])
```

Measure: Avg Satisfaction (used in Satisfaction by Segment)

```
Avg Satisfaction = AVERAGE(Sheet1[SatisfactionScore])
```

Charts used: Clustered Column (Spend by Age & Gender), Bar (Top Product Categories), Column (Satisfaction by Segment), Scatter (Purchases Vs Spend)

PAGE 3 — Churn Analysis

Measure: Total Customers (Churns by Segment)

```
Total Customers =  
CALCULATE(  
COUNTROWS(Sheet1)  
)
```

Measure: Active Customers

```
Active Customers =  
CALCULATE(  
COUNTROWS(Sheet1),  
Sheet1[ChurnStatus] = "No"  
)
```

Measure: Churn Rate (Churn Vs Active pie label)

```
Churn Rate =  
DIVIDE(  
COUNTROWS(FILTER(Sheet1, Sheet1[ChurnStatus] = "Yes")),  
COUNTROWS(Sheet1),  
0  
) * 100
```

Measure: Premium Customers

```
Premium Customers =  
CALCULATE(  
COUNTROWS(Sheet1),  
Sheet1[Segment] = "Premium"  
)
```

Measure: Premium Revenue

```
Premium Revenue =  
CALCULATE(  
SUM(Sheet1[TotalSpend]),  
Sheet1[Segment] = "Premium"  
)
```

Churned Customer Details Table — Filter applied in visual filter pane: Sheet1[ChurnStatus] = "Yes" Columns dragged: Name, Region, Segment, TotalSpend, SatisfactionScore

CALCULATED COLUMNS SUMMARY (New Column)

Calculated Column: Age Group

```
Age Group =  
SWITCH(  
TRUE(),  
Sheet1[Age] < 30, "18-29",  
Sheet1[Age] < 40, "30-39",  
Sheet1[Age] < 50, "40-49",  
"50+"  
)
```

Calculated Column: High Value Customer

```
High Value Customer =  
IF(Sheet1[TotalSpend] > 30000, "High Value", "Standard")
```

DATE TABLE + RELATIONSHIP

New Table: DateTable

```
DateTable = CALENDAR(  
MIN(Sheet1[LastPurchaseDate]),  
MAX(Sheet1[LastPurchaseDate])  
)
```

New Column inside DateTable: Month

```
Month = FORMAT(DateTable[Date], "MMMM")
```

New Column inside DateTable: Quarter

```
Quarter = "Q" & QUARTER(DateTable[Date])
```

New Column inside DateTable: Year

```
Year = YEAR(DateTable[Date])
```

Relationship: DateTable[Date] → Sheet1[LastPurchaseDate] Cardinality: Many to One (*:1) | Direction: Single

SLICERS (All Pages)

Slicer	Field to drag	Style
Product Category	Sheet1[ProductCategory]	Tile
Gender	Sheet1[Gender]	List
Segment	Sheet1[Segment]	List
Region	Sheet1[Region]	Dropdown